

## BANARAS HINDU UNIVERSITY

B.Sc. (Hons) — Semester II Examination, 2013-14 (Old Course)

## Paper No. MTB-202: Multivariable Calculus

Time: 3 Hours

Full Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Figures in right-hand margin indicate marks.*

**Question 1.**

- (a) Give the  $\varepsilon$ - $\delta$  definition of the limit of a function  $f : \mathbb{R}^2 \rightarrow \mathbb{R}$  as  $(x, y) \rightarrow (a, b)$ . Using the definition, show that:

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2}{x^2 + y^2}$$

does not exist.

[3]

- (b) Show that the simultaneous limit of  $f(x, y) = \frac{xy}{x^2 + y^2}$  does not exist at  $(0, 0)$ , but that the iterated limits exist and are equal.

[2]

- (c) Let  $f(x, y)$  be a homogeneous function of degree 5. Show that:

$$x^2 \frac{\partial^2 f}{\partial x^2} + 2xy \frac{\partial^2 f}{\partial x \partial y} + y^2 \frac{\partial^2 f}{\partial y^2} = 20f$$

[2]

- (d) State Taylor's theorem for functions of two variables.

[2]

- (e) Evaluate  $\int_0^\infty x^n e^{-x} dx$ .

[2]

- (f) Show that  $\int_0^\infty e^{-x^2} dy = \Gamma\left(\frac{n+1}{2}\right)$ ,  $n > 0$ .

[2]

- (g) Write a necessary condition for  $f(x, y)$  to have an extremal value.

[2]

- (h) If  $x, y, z$  are positive and  $h_1 \leq x + y + z \leq h_2$ , then find:

$$\iiint (x + y + z) x^m y^n z^l dx dy dz$$

[2]

- (i) If  $\vec{r}$  is a vector function of scalar  $t$  and  $\vec{a}$  is a constant vector, differentiate the following with respect to  $t$ :

(i)  $\vec{r} \cdot \vec{a}$

(ii)  $\vec{r} \times \vec{a}$

(iii)  $|\vec{r}| \frac{d\vec{r}}{dt}$

[3]

(j) Prove that  $\operatorname{div} \vec{r} = 3$ .

[2]

**Question 2.**

(a) Prove that the limit of the product of two functions of two variables equals the product of their limits. [7]

(b) Show that the function

$$f(x, y) = \begin{cases} \frac{xy}{\sqrt{x^2 + y^2}}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$$

is continuous at the origin.

[5]

**Question 3.**(a) If  $(a, b)$  is a point of the domain  $D \subset \mathbb{R}^2$  of a real-valued function  $f(x, y)$ , such that (i)  $f_x(a, b)$  exists, (ii)  $f_y(x, y)$  is continuous at  $(a, b)$ , then prove that  $f(x, y)$  is differentiable at  $(a, b)$ . [7](b) Show that  $f(x, y) = \sin x + \cos y$  is differentiable everywhere.

[5]

**Question 4.**(a) If  $A, B, C$  are the angles of a triangle  $ABC$ , find the maxima and minima of  $u = \sin A \cdot \sin B \cdot \sin C$ . [7](b) Transform the integral  $\int_0^a \int_0^{\sqrt{a^2 - x^2}} \sqrt{x^2 + y^2} dx dy$  by changing to polar coordinates and hence evaluate it. [5]**Question 5.**(a) Show that  $B(m, n) = \int_0^\infty \frac{y^{n-1}}{(1+y)^{m+n}} dy$ . [6](b) By changing the order of integration, evaluate  $\int_0^{\pi/2} \int_0^{2a \cos \theta} f(r, \theta) dr d\theta$ . [6]**Question 6.**(a) (i) If  $\vec{a}$  is a constant vector, find  $\operatorname{div}(\vec{r} \times \vec{a})$  and  $\operatorname{curl}(\vec{r} \times \vec{a})$ .

(ii) Prove that  $\text{grad}\left(\frac{1}{r}\right) = -\frac{\vec{r}}{r^3}$ .

[6]

(b) Prove that  $\text{grad}(\vec{a} \cdot \vec{b}) = \vec{a} \times \text{curl } \vec{b} + \vec{b} \times \text{curl } \vec{a} + (\vec{a} \cdot \text{grad})\vec{b} + (\vec{b} \cdot \text{grad})\vec{a}$ .

[6]

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**Question 7.**

(a) Verify Stokes' theorem for  $\vec{F} = (x^2 + y^2)\vec{i} - 2xy\vec{j}$  taken round the rectangle bounded by  $x = \pm a, y = 0, y = b$ .

[6]

(b) State and prove the Gauss Divergence Theorem.

[6]